

# ADVERSARIAL REASONING COUNCIL

## Standard Operating Procedure

Version 1.5 | **LOCKED** | Principal: Lanti

### 1. Purpose

This SOP governs the execution of all ARC sessions. It defines the workflow, role responsibilities, phase sequence, output standards, and Principal approval requirements for Low, Medium, and High risk tasks. ARC is a multi-model adversarial governance framework. Its purpose is to surface weaknesses, eliminate single-model bias, and produce verified, evidence-graded outputs before any recommendation enters an institutional system. The Principal holds final authority over all outputs. No ARC output is binding without Principal review and approval.

### 2. Scope

- All tasks classified LOW, MEDIUM, or HIGH risk per ARC Charter v1.5 Section 12.
- Manual sessions executed via web interfaces (Claude.ai, ChatGPT, Gemini).
- Automated sessions executed via the Stage 3 pipeline (arc\_pipeline.py with arc\_config.yaml).
- This SOP does not govern tasks outside the ARC framework or Principal personal research.

### 3. Council Structure

ARC operates with four functional roles. Default model assignments are determined by task type per Charter v1.5 Section 2.1. The Principal overrides any assignment using the ARC Brief Generator prior to session start.

| ROLE                   | DEFAULT MODEL     | RESPONSIBILITY                                                                                                             |
|------------------------|-------------------|----------------------------------------------------------------------------------------------------------------------------|
| GENERATOR              | Claude (default)  | Produces 3+ distinct candidate frameworks. Appends Assumption Log and Phase Confidence score.                              |
| RED TEAM               | Gemini (default)  | Generates independent solution first. Critiques all frameworks blind. Rates each weakness High, Medium, or Low.            |
| SYNTHESIZER / CHAIRMAN | Claude (default)  | Resolves sustained Red Team verdicts. Applies Evidence Taxonomy. Produces final synthesis.                                 |
| VERIFIER               | ChatGPT (default) | Classifies all claims: FACT, INFERENCE, SPECULATION, UNVERIFIED. Assigns Verification Confidence score. Flags if below 70. |

Default assignments by task type are codified in Charter v1.5 Section 2.1. The ARC Brief Generator auto-loads defaults and allows override before brief generation.

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## 4. Pre-Session Setup

Complete all steps before opening any model interface.

### 4.1 Principal Session Brief

The Principal Session Brief is the single source of truth for the session. It defines what the council knows, what it does not re-explain, and where harder critique is required.

- **Principal Name:**

*Your name. Appears in all three activation briefs.*

- **Task Type:**

*Select from the eight ARC task categories. Drives default role assignments.*

- **Risk Level:**

*LOW, MEDIUM, or HIGH. HIGH activates Phase 4C and requires PRIMARY source citations in Verification.*

- **Task:**

*State the task clearly. Include success criteria. Minimum two sentences.*

- **Domain Expertise:**

*What you bring to this session. Prevents the council from explaining what you already know.*

- **Current Baseline:**

*What you already know about this topic. Council does not re-cover it.*

- **Known Blind Spots:**

*Where you want harder critique. Where your certainty is lower.*

- **Active Task Context:**

*The project, decision, or initiative this session feeds.*

### 4.2 Generate Activation Briefs

Use the ARC Brief Generator at: <https://thorspeed-arc.github.io/arc-brief-generator/>

- Fill all fields in Panel 01 and Panel 02.
- Confirm role assignments match the task type. Override if needed.
- Click Generate Activation Briefs.
- Three briefs appear in Panel 03, tabbed by model: Claude, ChatGPT, Gemini.
- Copy each brief. Paste into the corresponding platform before providing any task input.

### 4.3 Model Confirmation

After pasting each activation brief, wait for the model to confirm its assigned role before providing the task. Do not proceed if confirmation is missing or the model states a different role.

## 5. Session Workflow

The ARC workflow consists of nine phases. Phases 1 and 2 are completed by the Principal before council engagement. Phases 3 through the Decision Log are executed by council models under Principal oversight.

PRE-SESSION (Principal Only)

PHASE 1 — Task Definition: Principal defines task, risk level, success criteria, and session brief.  
 PHASE 2 — Brief Distribution: Principal generates activation briefs and distributes to all three model platforms. Waits for role confirmation from each model.

| PHASE    | NAME                   | ACTIONS                                                                                                                                                                                                                                                                                                                                                    |
|----------|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PHASE 3  | Generation             | Generator model produces 3+ distinct candidate frameworks independently. Each framework must be meaningfully different in approach. Generator appends an Assumption Log (A1, A2... each marked Untested) and a Phase Confidence score (0-100) with one-sentence rationale.                                                                                 |
| PHASE 4  | Red Team Critique      | Red Team model generates its own independent solution first (labeled INDEPENDENT SOLUTION), then critiques all Generator frameworks blind without knowing which model produced them. Each weakness is rated High, Medium, or Low. Every Assumption Log item is challenged: Challenged, Confirmed, or Invalidated. Red Team appends Phase Confidence score. |
| PHASE 4B | Rebuttal               | Generator model reviews Red Team critique and defends its logic. Corrects genuine misunderstandings. Offers solutions to valid critiques. Input is token-limited to prevent padding. Generator appends updated Phase Confidence score.                                                                                                                     |
| PHASE 4C | Final Red Team Verdict | HIGH RISK ONLY. Red Team issues a binding verdict per weakness after reviewing Generator rebuttal: SUSTAINED, PARTIALLY SUSTAINED, or WITHDRAWN. One sentence of reasoning per verdict required. This phase does not exist for LOW or MEDIUM risk sessions.                                                                                                |
| PHASE 5  | Synthesis              | Synthesizer / Chairman reads all prior phases. Resolves all Sustained and Partially Sustained verdicts. Carries forward only Confirmed assumptions. Applies Evidence Taxonomy to all material claims (FACT, INFERENCE, SPECULATION, UNVERIFIED). Produces consolidated output. Input is token-limited.                                                     |
| PHASE 6  | Verification           | Verifier model classifies every material claim in the synthesis output. For HIGH risk: cites a primary source for every FACT-rated claim. Assigns Verification Confidence score (0-100). If below 70: names the specific claims lowering the score. If below 50: states RECOMMEND FULL RESTART with explanation.                                           |
| PHASE 6B | Ensemble Ranking       | All three models score the verified output independently on three dimensions (0-100 each): Technical Accuracy, Completeness, Risk Coverage. Scores are recorded in the Decision Log. Models score the content, not the model that produced it.                                                                                                             |
| REVIEW   | Principal Approval     | Principal reviews Verification output against original Session Brief. For HIGH risk tasks: explicit Principal sign-off required before output enters any institutional system. Principal retains final authority and override rights at all times.                                                                                                         |
| END      | Decision Log           | Final recommendation recorded. Log includes: reasoning chain, evidence classification, risk level, Phase Confidence scores, Verification Confidence, Ensemble Ranking scores, and any sustained weaknesses. Output enters institutional system after Principal approval.                                                                                   |

## 6. Risk Level Differences

Risk level determines which phases are active and what standards apply to Verification.

| REQUIREMENT                   | LOW         | MEDIUM      | HIGH                  |
|-------------------------------|-------------|-------------|-----------------------|
| Phase 4C Active               | No          | No          | Yes                   |
| Primary source citations      | No          | No          | Yes (all FACT claims) |
| Verification Confidence floor | 50          | 60          | 70                    |
| Principal sign-off required   | Recommended | Recommended | Required              |
| Automated pipeline required   | Optional    | Optional    | Required (Rule 11)    |
| Anonymity Rule active         | No          | No          | Yes (Rule 12)         |

## 7. Output Standards

### 7.1 Evidence Taxonomy

All Synthesizer and Verifier outputs must classify every material claim using the ARC Evidence Taxonomy:

- **FACT** — Verifiable via primary source. Verifier must cite source on HIGH risk tasks.
- **INFERENCE** — Logical conclusion from established facts. Source not required but must be stated.
- **SPECULATION** — Plausible but not grounded in established evidence. Must be flagged explicitly.
- **UNVERIFIED** — Claim present in output but not classified. Verifier flags all instances.

### 7.2 Phase Confidence

Every model appends a Phase Confidence score (0-100) with a one-sentence rationale at the end of each phase output. Scores below 70 require a named explanation. Scores below 50 trigger a Principal escalation decision.

### 7.3 Verification Confidence

The Verifier assigns a single Verification Confidence score to the full synthesis output. Thresholds by risk level are defined in Section 6. If the score falls below the floor, the Verifier names the specific claims driving the reduction. If below 50, the Verifier states RECOMMEND FULL RESTART.

### 7.4 Decision Log

Every completed session produces a Decision Log entry containing:

- Final recommendation.
- Evidence classification for all material claims.
- Risk level and task type.
- Phase Confidence scores per phase per model.
- Verification Confidence score.
- Ensemble Ranking scores (Phase 6B).
- All Sustained or Partially Sustained Red Team verdicts with disposition.
- Principal approval status.

## 8. Principal Authority

The Principal is the sole decision authority in ARC. The council produces outputs. The Principal decides.

- The Principal overrides any council output at any phase without justification.
- The Principal terminates any session at any phase without justification.
- The Principal reassigns any model role prior to brief generation.
- HIGH risk task outputs do not enter any institutional system without explicit Principal sign-off.
- The Principal is not bound by Verification Confidence scores. The Principal decides whether to proceed, restart, or override.

## 9. Operational Learning

Per Charter v1.5 Section 20, ARC revisions require operational evidence. Design-phase iteration ended at v1.5. The following metrics are tracked per session and reviewed after every ten sessions.

| METRIC                             | WHAT IT MEASURES                                           | REVISION TRIGGER                       |
|------------------------------------|------------------------------------------------------------|----------------------------------------|
| <b>Session Friction</b>            | Principal-rated 1-5 per session. Measures workflow burden. | Above 3 consistently                   |
| <b>Yield</b>                       | ARC output quality vs single-model baseline.               | Below 70% across 10 sessions           |
| <b>Phase Failure Concentration</b> | Which phase fails most frequently by task type.            | 3+ failures same phase, same task type |
| <b>False Rigor</b>                 | Process appears disciplined but adds no quality.           | 5+ consecutive sessions                |
| <b>Role Fit</b>                    | Model performance per assigned role per session.           | 10 sessions same task type             |

## 10. Session Quick Reference

Use this checklist at the start of every ARC session.

### PRE-SESSION CHECKLIST

- [ ] Open ARC Brief Generator: <https://thorspeed-arc.github.io/arc-brief-generator/>
- [ ] Fill Panel 01: Principal name, task type, risk level, task, role assignments.
- [ ] Fill Panel 02: Domain expertise, baseline, blind spots, task context, output style.
- [ ] Click Generate Activation Briefs.
- [ ] Copy Claude brief → paste into Claude.ai → wait for role confirmation.
- [ ] Copy ChatGPT brief → paste into ChatGPT → wait for role confirmation.
- [ ] Copy Gemini brief → paste into Gemini → wait for role confirmation.
- [ ] Provide task to Generator model only. Do not share task with other models yet.

### SESSION EXECUTION CHECKLIST

- [ ] PHASE 3: Generator produces 3+ candidate frameworks with Assumption Log.
- [ ] PHASE 4: Red Team critiques blind. Rates weaknesses. Challenges assumptions.
- [ ] PHASE 4B: Generator rebuts Red Team critique.

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- [ ] PHASE 4C: (HIGH RISK ONLY) Red Team issues SUSTAINED / PARTIALLY SUSTAINED / WITHDRAWN verdicts.
  - [ ] PHASE 5: Synthesizer resolves verdicts. Applies Evidence Taxonomy.
  - [ ] PHASE 6: Verifier classifies all claims. Assigns Verification Confidence score.
  - [ ] PHASE 6B: All three models score output: Technical Accuracy, Completeness, Risk Coverage.
  - [ ] REVIEW: Principal reviews output against original Session Brief.
  - [ ] END: Record Decision Log entry. Sign off if HIGH risk before entering institutional system.
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ARC Charter v1.5 — LOCKED | This SOP supersedes all prior versions | Amendments require operational evidence per Section 20

Brief Generator: <https://thorspeed-arc.github.io/arc-brief-generator/>